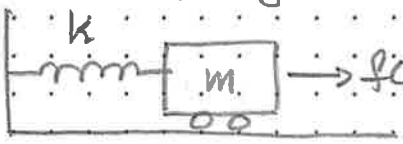
EXAMPLE 4 Solve $y'' + 16y = e^{3x}$

using the ERF approach to find the particular solution. Write the down the complete solution.

Solution: Equivalent spring-mass system for the given ODE:



$\left[\frac{k}{m} = 16 \ N/m\right]$ $\left[m = 1 \ kg\right]$ with appropriate SI units.

Homogeneous solution: (y_h)

Homogeneous equation for ODE: $[y'' + 16y = 0]$

Characterostic equation: $s^2 + 16 = 0$

$s = -\sqrt{16} \Rightarrow \boxed{s_1 = 4i} \quad \boxed{s_2 = -4i}$

Homogeneous solution:

$$y_h = C_1 e^{s_1 x} + C_2 e^{s_2 x}$$

$$\Rightarrow \boxed{y_h = C_1 e^{4ix} + C_2 e^{-4ix}}$$

$y_h = C_1[\cos(4x) + i\sin(4x)] + C_2[\cos(4x) - i\sin(4x)]$

$\Rightarrow y_h = (C_1 + C_2)\cos(4x) + (C_1 - C_2)i\sin(4x)$

$\Rightarrow$ Using theorem for complex solution,

$$\boxed{y_h = C_1\cos(4x) + C_2\sin(4x)}$$

Particular solution: (y_p)

$y_p'' + 16y_p'' = e^{3x}$ where y_p is the particular solution

Express forcing term $f(x) = e^{3x}$ as $e^{(a+i\omega)x}$:

$e^{3x} = \mathrm{Re}\{e^{(3+i\omega)x}\}$ where $\omega = 0$

Complexify the ODE:

$y_p'' + 16y_p = e^{3x} \Rightarrow z_p'' + 16z_p = e^{(3+i\omega)x}$ where $\omega = 0$.

$[y_p = z_p$, where $\omega = 0]$

Write down the particular solution:

$$z_p = \frac{\mathrm{Re}\{e^{(3+i\omega)x}\}}{p(3+i\omega)} \text{ where } \omega = 0$$

$$\Rightarrow z_p = \frac{e^{3x}}{p(3)} = \frac{e^{3x}}{3^2+16} \quad \{\text{Recall: } p(s) = s^2 + 16\}$$

$$\Rightarrow z_p = \frac{e^{3x}}{25}$$

The particular solution:

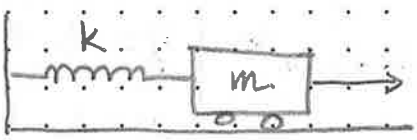
$$\boxed{y_p = \frac{e^{3x}}{25}}$$

Complete solution: $y(x) = y_p + y_h \Rightarrow \boxed{y(x) = \left(\frac{e^{3x}}{25}\right) + C_1\cos(4x) + C_2\sin(4x)}$

EXAMPLE 5 Solve $y'' - 4y = \sinh(x)$

using the ERF approach. Write down the complete solution.

Solution: Equivalent spring-mass system for the given ODE:



$\left[\begin{array}{ll} \frac{k}{m} = 4 & N/m^2 \\ m = 1 & kg \end{array}\right]$ in appropriate SI units

Homogeneous solution: (y_h)

$y'' - 4y = 0$

Characteristic equation:

$s^2 - 4 = 0 \Rightarrow s = \sqrt{2} = \pm 2$

[Let $p(s) = s^2 - 4$] $\Rightarrow$ $s_1 = 2$ $s_2 = -2$

Homogeneous solution:

$y_h = C_1 e^{s_1 x} + C_2 e^{s_2 x}$

$\Rightarrow \boxed{y_h = C_1 e^{2x} + C_2 e^{-2x}}$

Particular solution: (y_p)

$y_p'' - 4y_p = \sinh(x)$

Express the forcing term, $f(x) = \sinh(x)$ as $e^{\alpha t}$:

$\sinh x = \left(\frac{e^x}{2} - \frac{e^{-x}}{2}\right) = e^{\alpha t}$

Complexify the ODE:

$y_p'' - 4y_p = \sinh x$ | $z_p = y_p$

$\Rightarrow z_p'' - 4z_p = \frac{e^x}{2} - \frac{e^{-x}}{2}$

$\Rightarrow z_{p1}'' - 4z_{p1} = \frac{e^x}{2}$ $\quad z_{p2}'' - 4z_{p2} = -\frac{e^{-x}}{2}$

Write down the particular solution:

$z_{p1} = \frac{\frac{1}{2}e^x}{p(1)}$ | Recall $p(s) = s^2 - 4$

$z_{p1} = \frac{\frac{1}{2}e^x}{1^2 - 4} = -\frac{1}{6}e^x$

Write down the particular solution:

$z_{p2} = \frac{-\frac{1}{2}e^{-x}}{p(-1)}$

$z_{p2} = \frac{-\frac{1}{2}e^{-x}}{(-1)^2 - 4} = \frac{1}{6}e^{-x}$

$\Rightarrow \boxed{z_p = z_{p1} + z_{p2}}$ → See Aside ②

$\Rightarrow z_p = -\frac{1}{6}e^x + \frac{1}{6}e^{-x}$

$\Rightarrow \boxed{y_p = -\frac{1}{6}e^x + \frac{1}{6}e^{-x}}$

① **Aside:** $\left[\sinh x = \frac{e^x - e^{-x}}{2}\right]$

This the trigonometric formula for $\sinh x$ in terms of an exponential.

② **Aside:** $z_p'' - 4z_p = \frac{e^x}{2} - \frac{e^{-x}}{2}$

$\Rightarrow \left[z_{p1}'' - 4z_{p1} = \frac{e^x}{2}\right]$ & $\left[z_{p2}'' - 4z_{p2} = -\frac{e^{-x}}{2}\right]$

By principle of superposition we can solve for z_{p1} & z_{p2} & then sum the two solutions; i.e.; $z_p = z_{p1} + z_{p2}$

Complete solution: $\boxed{y(t) = y_p + y_h}$ $\Rightarrow y(t) = \left(-\frac{1}{6}e^x + \frac{1}{6}e^{-x}\right) + \left(C_1 e^{2x} + C_2 e^{-2x}\right)$

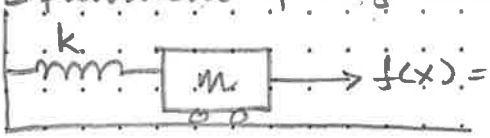
$\Rightarrow y(t) = -\frac{1}{3}\left(\frac{e^x + e^{-x}}{2}\right) + \left(C_1 e^{2x} + C_2 e^{-2x}\right)$

$\Rightarrow \boxed{y(t) = -\frac{1}{3}\sinh(x) + C_1 e^{2x} + C_2 e^{-2x}}$

EXAMPLE 6 Solve $y'' + 9y = \sin(2x)$

using ERF approach. Write down the complete solution for the initial states $y(0) = 1$; $y'(0) = -1$

Solution: Equivalent spring-mass system for the given ODE



$\left[\frac{k}{m} = 9\ N/m^2,\ m = 1\ kg\right]$ in appropriate SI units.

Homogeneous solution: (y_h)

$y'' + 9y = 0$

Characteristic equation:

$s^2 + 9 = 0$ Let $p(s) = s^2 + 9$

$s = \pm\sqrt{-9} \Rightarrow \boxed{s_1 = 3i}\ \boxed{s_2 = -3i}$

Homogeneous solution:

$y_h = c_1 e^{s_1 x} + c_2 e^{-s_2 x}$

$\Rightarrow y_h = c_1 e^{3ix} + c_2 e^{-3ix}$

$\Rightarrow y_h = c_1[\cos(3x) + i\sin(3x)] + c_2[\cos(3x) - i\sin(3x)]$

$= (c_1 + c_2)\cos(3x) + i(c_1 - c_2)\sin(3x)$

Using the theorem of complex solutions

$\Rightarrow \boxed{y_h = c_1\cos(3x) + c_2\sin(3x)}$

Particular solution: (y_p)

$y_p'' + 9y_p = \sin(2x)$

Express the forcing term as $e^{(a+i\omega)x}$.

$\sin(2x) = Im\{e^{(a+2i)x}\}$ where $a = 0$

$= Im\{e^{i2x}\}$

complexify the ODE:

$y_p'' + 9y_p = \sin(2x) \Rightarrow z_p'' + 9z_p = e^{i2x}$

$\because y_p = Im\{z_p\}$ solves $y_p'' + 9y_p = \sin(2x)$.

Write down the particular solution:

$\left[z_p = \frac{e^{(a+i\omega)x}}{p(a+i\omega)}\right]$ Recall: $p(s) = s^2 + 9$

$\Rightarrow z_p = \frac{e^{i2x}}{p(2i)} = \frac{e^{i2x}}{(2i)^2 + 9}$

$\Rightarrow z_p = \frac{e^{i2x}}{-4 + 9} = \frac{e^{i2x}}{5}$

$\boxed{z_p = \frac{e^{i2x}}{5}}$

The particular solution: $y_p = Im\{z_p\}$

$y_p = Im\left\{\frac{e^{i2x}}{5}\right\} = \left(\frac{1}{5}\right) Im\{\cos(2x) + i\sin(2x)\}$

$\boxed{y_p = \left(\frac{1}{5}\right)\sin(2x)}$

Complete solution: $y(t) = y_p + y_h$

$\boxed{y(t) = \left(\frac{1}{5}\right)\sin(2x) + c_1\cos(3x) + c_2\sin(3x)}$

Apply initial states:

$y(0) = \left(\frac{1}{5}\right)\sin(2\times 0) + c_1\cos(3\times 0) + c_2\sin(3\times 0) = 1$

$\Rightarrow \boxed{c_1 = 1}$

$y'(t) = \left(\frac{1}{5}\right)(2)\cos(2x) + c_1(3)\sin(3x) + c_2(3)\cos(3x)$

$y'(0) = \left(\frac{1}{5}\right)(2)\cos(2\times 0) - c_1(3)\sin(3\times 0) + c_2(3)\cos(3\times 0) = -1$

$3c_2 + \frac{2}{5} = -1 \Rightarrow \boxed{c_2 = -\frac{7}{15}}$

Final solution:

$$y(t) = \frac{1}{5}\sin(2x) + \cos(3x) - \left(\frac{7}{15}\right)\sin(3x).$$